

Signals and Systems

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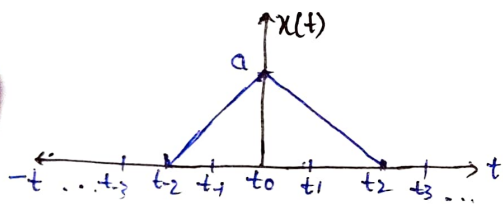
Lecture-01

* Signals and their responses

- Signal - Signal is basically a function of one or more independent variables that describes a physical phenomenon.
- The input signal then passes through a system governed by some physical phenomena.
- The system responds to the input signal resulting in some interesting signal characteristics at the output.
- Signal varies with time but system response is governed by frequency characteristics.

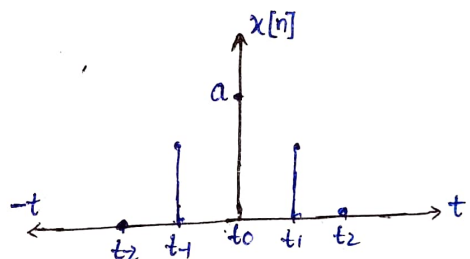
→ Continuous Time Signals

These signals are specified for every value of time (t).



→ Discrete Time Signals

specified at discrete time intervals



$$\Delta t = t_1 - t_0 = t_2 - t_1 = t_3 - t_2 = t_n - t_{n-1}$$

$$\Delta t = T$$

* Signal Energy and Power

$$\text{Power} = \int_{t_1}^{t_2} x^2(t) dt$$

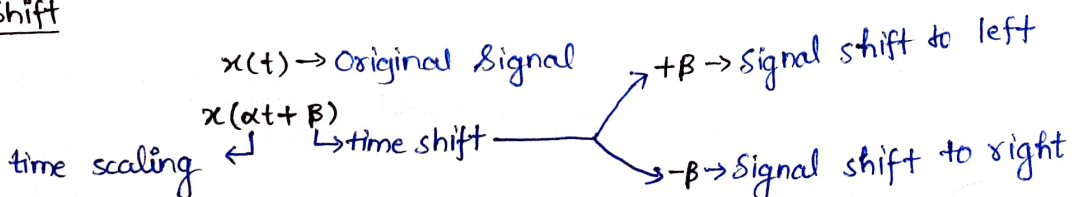
Signal represented as where, $x(t) \rightarrow$ fn of time.

Note: The eg. expression is dimensionless, meaning it is not the actual physical energy, but it's a mathematical expression that behaves like energy.

eg. Electrical Energy $\rightarrow \frac{1}{R} \int_{t_1}^{t_2} v^2(t) dt$
 $\left[\frac{v^2(t)}{R} \right]$ is the power dissipated, integrating it over time we get energy.

→ In any system, it is important to find the independent variable with respect to signal varies.

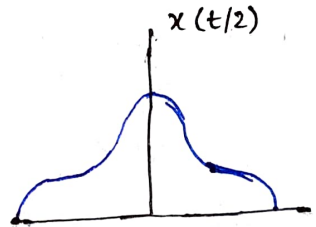
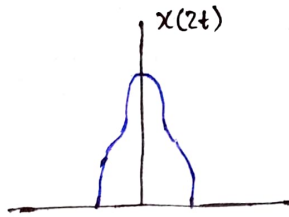
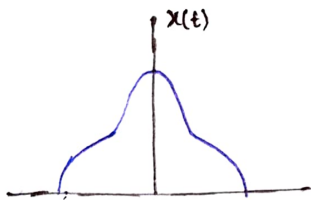
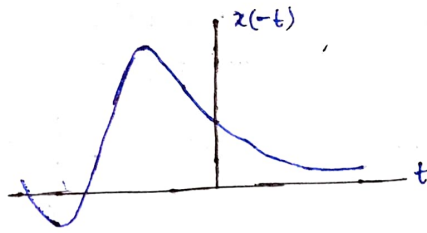
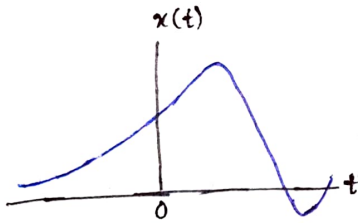
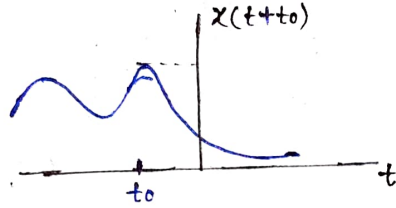
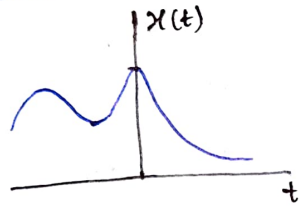
* Time Shift



→ $x(-t+\beta)$: time reversed version of $x(t+\beta)$ obtained by reflecting graph about time axis.
 $y \leftarrow \text{wrong}$

* Time Scaling →

- The signal is linearly stretched if $|a| < 1$
- Compressed when $|a| > 1$.
- Reversed in time $a < 0$



* Periodicity of Signals

- if signal has a periodicity T then, $x(t+T) = x(t)$
- Now, $x(t+mT) = x(t)$, the signal is periodic for $m=1,2,3,\dots$
- and the signal is periodic for $T, 2T, 3T$.
- the smallest T for which equations hold is called the fundamental period and rest its multiples are harmonics.
- Analogously, discrete time signals can have period N so that

$$x[n] = x[n+N]$$

Note: There is risk in determining periodicity if the signal is discontinuous at some point.

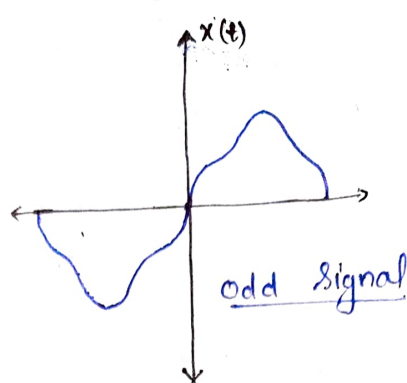
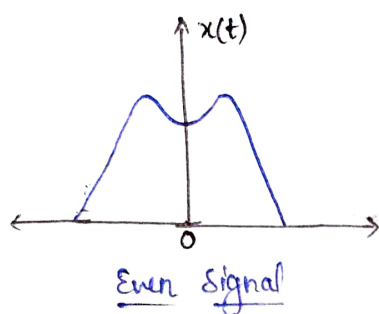
* Even and Odd Signals

- Signal is even, if $x(-t) = x(t)$ → for continuous
 $x[-n] = x[n]$ → for discrete
- Signal is odd, if $x(-t) = -x(t)$ → for continuous
 $x[-n] = -x[n]$ → for discrete

→ Every signal can be split into even and odd signal.

$$x_{\text{even}} = \frac{1}{2} [x(t) + x(-t)]$$

$$x_{\text{odd}} = \frac{1}{2} [x(t) - x(-t)]$$



* Real and Exponential Signals

→ Continuous time complex exponential signals:

$$x(t) = Ce^{at}$$

where, $C, a \rightarrow$ complex no.

→ When C, a are purely real i.e., imaginary part is absent then,

for +ve $a \rightarrow$ growing exponential

for -ve $a \rightarrow$ decaying exponential

→ When a is purely imaginary, $x(t) = e^{j\omega t}$

$$e^{j\omega t} = e^{j\omega(t+T)}, \text{ Period} = T$$

→ Periodicity : for $\omega_0 = 1$, Period = T

otherwise, Period $T_0 = \frac{2\pi}{|\omega_0|}$

→ $e^{j\omega t}$ and $e^{-j\omega t}$ have same fundamental period.

→ $\omega_0 = 2\pi f_0$, $f_0 \rightarrow$ frequency

→ Using Euler results, $x(t) = e^{j\omega t} = \cos \omega t + j \sin \omega t$.

→ Therefore the sinusoidal signal can be expressed as,

$$A \cos(\omega t + \phi) = \frac{A}{2} e^{j(\omega t + \phi)} + \frac{A}{2} e^{-j(\omega t + \phi)}$$

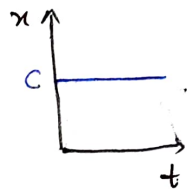
SIGNAL AND SYSTEM

* Energy of Sinusoidal Signals

→ Sinusoids describe many physical phenomenon like current oscillation of LC circuit, angular position in simple pendulum.

→ Special Case: Constant Signal → If signal is constant i.e., it doesn't oscillate $\Rightarrow \omega_0 = 0$. The period T_0 becomes undefined (bcz a constant signal doesn't repeat in cycles), its frequency is 0.

eg. $x(t) = C$



→ For periodic signals like sinusoids, total ~~power~~ energy is infinite bcz the oscillations never dies out. - keeps going forever.
→ The avg. power is finite however.

→ Energy in one period, for complex sinusoid (say, $e^{j\omega_0 t}$)

$$E = \underbrace{\int_{-\infty}^{\infty} |x(t)|^2 dt}_{\text{Total energy for a signal}} \Rightarrow E = \int_0^{T_0} |e^{j\omega_0 t}|^2 dt \Rightarrow E = \int_0^{T_0} 1 \cdot dt = T_0 \quad \left[\begin{array}{l} \text{By Euler's formula} \\ |e^{j\omega_0 t}| = 1 \end{array} \right]$$

→ Average Power
 $P = \frac{E}{T_0} = \frac{T_0}{T_0} = 1$, so a sinusoid has unit average power.

→ If extending this to multiple periods, the average power remains 1 but Energy $\rightarrow \infty$.

Hence, Sinusoids have finite power, infinite energy hence they're Power Signals.

→ Harmonically related sinusoids,

$$\phi_k(t) = e^{jk\omega_0 t}; \quad k = 0, \pm 1, \pm 2$$

→ General Complex Signal = Sinusoidal $\times$ growing or decaying exponential.

$$c = x(t) e^{-at}$$

$$x(t) = C e^{at}$$

$$x(t) = C e^{(\sigma + j\omega_0)t} = C e^{\sigma t} \cdot e^{j\omega_0 t}$$

where,

C = complex constant (amp & phase)

$$a = \sigma + j\omega_0$$

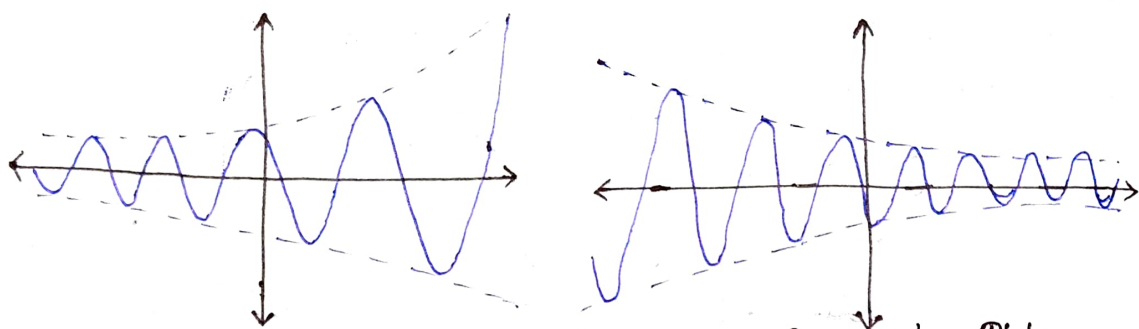
ω_0 $\rightarrow$ angular freq.

$e^{\sigma t}$ $\rightarrow$ amplitude factor
 $\rightarrow$ expo. growth if $\sigma > 0$, decay if $\sigma < 0$ $\rightarrow$ controls growth/decay.

$e^{j\omega t} \rightarrow \cos(\omega t) + j\sin(\omega t) \rightarrow$ oscillatory part
 $\rightarrow$ Signal is basically a sinusoid with exponentially changing amplitude.

Role of C : $C = |C|e^{j\theta}$ where $|C| = \text{magnitude}$
 $\theta = \text{phase shift}$

Thus, $x(t) = |C|e^{\alpha t}e^{j(\omega t + \phi)}$ $\rightarrow$ Exponential envelop $e^{\alpha t}$
 Oscillating sinusoid, freq ω .
 Phase shift ϕ
 Amplitude scaling $|C|$.



Continuous Periodic Growing & Decaying Plots

* Discrete time complex signal

$$x[n] = C\alpha^n = |C||\alpha|^n e^{j(\omega n + \theta)} \quad (\text{instead of } t \text{ we have discrete } n \text{ values})$$

$$x[n] = |C||\alpha|^n \cos(\omega n + \theta) + j|C||\alpha|^n \sin(\omega n + \theta)$$

$|\alpha| = 1 \rightarrow$ Amplitude is constant $\rightarrow$ signal is sinusoid

$|\alpha| < 1 \rightarrow$ The sinusoid is multiplied by a decaying exponential envelop.

$|\alpha| > 1 \rightarrow$ The sinusoid is multiplied by a growing exponential envelop.

$\rightarrow$ Periodicity

$$x[n] = e^{j\omega_0 n}$$

$$\text{Now if } e^{j(\omega_0 + 2\pi k)n} = e^{j\omega_0 n} \cdot e^{j \cdot 2\pi k n}, \quad k \in \mathbb{Z}$$

$$\text{But } e^{j2\pi k n} = \cos(2\pi k n) + j\sin(2\pi k n) = 1$$

$$\text{So, } e^{j(\omega_0 + 2\pi k)n} = e^{j\omega_0 n}$$

Thus, discrete-time freq. repeat every 2π hence,

$$0 \leq \omega_0 < 2\pi \quad \text{or} \quad -\pi \leq \omega_0 < \pi$$

Condition for Periodicity: $e^{j\omega_0(n+N)} = e^{j\omega_0 n}$
 $e^{j\omega_0 N} = 1$

That happens when,

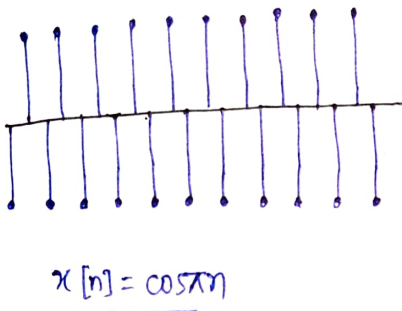
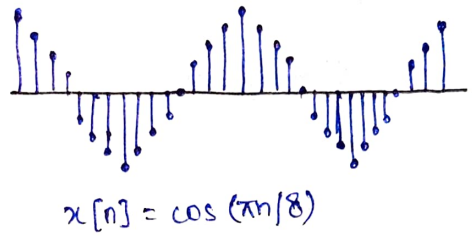
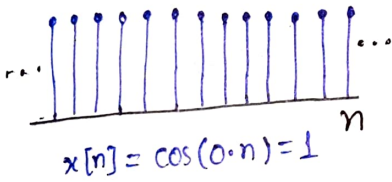
$$\omega_0 N = 2\pi m \quad \text{where } m \in \mathbb{Z}$$

$$\frac{\omega_0}{2\pi} = \frac{m}{N} \Rightarrow N = \frac{2\pi m}{\omega_0}$$

Note: The smallest N satisfying this above condⁿ is called fundamental period.

eg. if $\omega_0 = \frac{\pi}{2} \Rightarrow N = \frac{2\pi m}{\pi/2} \Rightarrow N = 4$ ($m=1$) is fundamental period.

$$\text{if } \omega_0 = 0.1\pi \Rightarrow N = \frac{2\pi \times 1}{0.1\pi} \Rightarrow \underline{N=20}$$



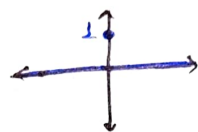
* Periodicity Properties Comparison

Property	Continuous Time Signal	Discrete Time Signal
Frequency Distinctness	Distinct for every different ω_0	Signal repeats every 2π i.e., identical in every 2π interval
Periodicity Condition	Always periodic for any ω_0	Periodic only if $\omega_0 = \frac{2\pi m}{N}$ where $m, N \in \mathbb{Z}$
Fundamental Frequency	ω_0	$\frac{\omega_0}{m}$
Fundamental Period	$T = \frac{2\pi}{\omega_0}$	$N = \frac{2\pi m}{\omega_0}$ i.e., a multiple of continuous time period,

* Impulse and step-up signal

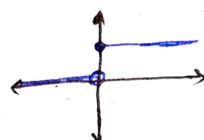
1. Impulse signal $\delta[n]$

$$\delta[n] = \begin{cases} 1, & n=0 \\ 0, & n \neq 0 \end{cases}$$



2. Unit step signal $u[n]$

$$u[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$$



→ Relation b/w $\delta[n]$ & $u[n]$: $\delta[n] = u[n] - u[n-1]$

$$u[n] = \sum_{m=-\infty}^n \delta[m]$$

→ For continuous time,

$$\delta(t) = \frac{d}{dt} u(t)$$

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

After integration τ disappears only t remains
 $\tau \rightarrow$ dummy variable for integration

★ System → In signal processing, a system is anything that takes an input signal and produces an output signal according to some rule.

$$x(t) \xrightarrow{\text{System}} y(t)$$

eg. of system → filter, amplifier, delay, differentiator, etc.

* Continuous Time Systems - Input signal is a fn of continuous time t , and the output is also continuous in time.
 eg. an analog circuit, mechanical system, etc.

$$x(t) \xrightarrow{\text{System}} y(t)$$

eg. RC circuit

Input source → Voltage $V_s(t)$

Output: Capacitor Voltage $V_c(t)$

System Relationship

$$\frac{dV_c(t)}{dt} + \frac{1}{RC} V_c(t) = \frac{1}{RC} V_s(t)$$

eg. Vehicle Dynamics

Input: Applied force $F(t)$

Output: Velocity $v(t)$ of the vehicle

$$\text{Eqn: } \frac{dv}{dt} + \frac{P}{m} \cdot v(t) = \frac{1}{m} F(t)$$

where $P \rightarrow$ friction coefficient
 $m \rightarrow$ mass.

* General Form

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

where, $x(t)$ = input
 $y(t)$ = output
 $a, b \rightarrow$ cons. depend on system properties,

* Discrete Time Systems :- Input signal is discrete-time, and output is also discrete time.

$$x[n] \xrightarrow{\text{System}} y[n]$$

eg. 1) Bank Balance

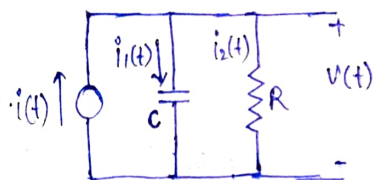
$$y[n] = (1+p)y[n-1] + x[n]$$

$\leftarrow$ balance at month n $\downarrow$ interest rate per month $\leftarrow$ balance from prev month $\rightarrow$ net deposit made in month n

Current balance = (Prev Bal + Interest) + new deposit

$\rightarrow$ General Form $\rightarrow y[n] + ay[n-1] = bx[n]$

* Interconnection of Systems



$\rightarrow$ (a) Capacitor Subsystems

$$v(t) = \frac{1}{C} \int_{-\infty}^t i_2(\tau) d\tau$$

Input: Capacitor Current ($i_1(t)$)
 Output: Capacitor Voltage ($v(t)$)

$\rightarrow$ (b) Resistor Subsystems

$$i_2(t) = \frac{v(t)}{R}$$

$\rightarrow$ (c) Interconnection

• At the summing junction
 $i(t) = i_1(t) + i_2(t)$

$\rightarrow$ Thus, any complicated circuit/system can be decomposed into subsystem

* Memoryless and Systems with memory

1. Memoryless System $\rightarrow$ Output depends only on present input, not on past or future values.

eg: $i(t) = \frac{v(t)}{R}$ (Ohm's law)

2. Systems with memory $\rightarrow$ Output depends on the present input as well as past inputs. eg. Capacitor $v(t) = \frac{1}{C} \int i(t) dt$, Voltage depends on how much charge is accumulated.

* Invertibility of Systems

→ Distinct inputs leads to distinct outputs

→ Inverse System

Suppose you have a system S with input $x(t)$ and output $y(t)$

$$y(t) = S[x(t)]$$

Now if there exists another system S^{-1} (inverse) such that when you pass $y(t)$ through it you get back $x(t)$.

$$S^{-1}[y(t)] = x(t)$$

then system S is invertible.

Cascaded view: $x(t) \xrightarrow{S} y(t) \xrightarrow{S^{-1}} x(t)$

eg. System: $y(t) = 2x(t)$, Inverse: $x(t) = \frac{y(t)}{2}$

eg. of non-invertible system (even power)

$y(t) = x^2(t) \rightarrow$ Now we get x^2 both ± 2 hence input is not unique.

Note: System: $y(t) = 0$, is invertible for all inputs.

* Casuality of Systems

→ A system is casual if the output at any time depends only on the present and past inputs - never on future inputs.

→ If $x_1(t) = x_2(t)$ for all $t \leq t_0$, then the outputs are also same for all $t \leq t_0$.

eg. i) A ohmic resistor (Ohm's law $V=IR$)

ii) An RC Circuit $\rightarrow$ Depends on present + past values of input voltage.

→ eg(s) of non casual systems include,

i) Image Processing

ii) Meteorology

(iii) Stock Market

* Stability of Systems

→ A system is stable if small/bounded inputs produce bounded outputs. This is called BIBO stability.

eg. i) Pendulum

→ Normal Pendulum is a stable system.

→ Inverted Pendulum (stick upright) is unstable.

→ for unstable system output grows uncontrollably for small/bounded inputs.

eg. In electrical systems, resistance turns energy into heat.
In mechanical system, friction or drag dissipates energy.

Thus stability often comes from energy dissipation.
This prevents output from diverging forever.

* Time Invariance of Systems

→ A system is time-invariant if shifting the input signal in time results in the same shift in the output, without changing its shape.

→ System Parameters do not change with time.

→ System followed by delay = delay followed by system.

eg. Consider system $c(t) = f(x(t)) = t \cdot x(t)$.

Delaying by input $x(t-\tau)$

Case 1) Delay after system

$$c_1 = \text{delay}(f(x(t))) = t \cdot x(t-\tau)$$

Case 2) Delay before system.

$$c_2 = f(x(t-\tau)) = (t-\tau) \cdot x(t-\tau)$$

$\therefore c_1 \neq c_2$; the system is not time invariant.

→ Condition for Time Invariance

$$x(t-\tau) \rightarrow y(t-\tau)$$

Input shifts by τ , output must also shift by the same τ .

→ eg. of time varying system,

$$y(t) = x(t) + 3$$

→ With delayed input, $y_1(t) = x(t-\tau) + 3$

→ With delayed output, $y_2(t) = y(t-\tau) = x(t-\tau) + 3$

$y_1 = y_2 \rightarrow$ Time-Invariant!

→ eg. of non-time invariant system,

$$y(t) = t x(t)$$

→ With delayed input; $y_1(t) = t \cdot x(t-\tau)$

→ With delayed output: $y_2(t) = y(t-\tau) = (t-\tau) x(t-\tau)$

$y_1 \neq y_2 \rightarrow$ not Time Varying.

* Linearity of Systems

→ A system is linear if it satisfies the Superposition Principle:

$$T\{a_1 x_1(t) + a_2 x_2(t)\} = a_1 T\{x_1(t)\} + a_2 T\{x_2(t)\} \quad \text{where } T: \text{system operator,}$$

i.e., → If input is $x_1(t)$, output is $C_1(t)$

→ If input is $x_2(t)$ output is $C_2(t)$

→ Then for input $a_1 x_1(t) + a_2 x_2(t)$, output must be $a_1 C_1(t) + a_2 C_2(t)$

eg. $T\{x(t)\} = 2x(t)$

Picking 2 input signal $x_1(t) = t$ $x_2(t) = \sin t$

Outputs: $C_1(t) = T\{x_1(t)\} = 2t$

$$C_2(t) = T\{x_2(t)\} = 2\sin t$$

Taking a L.C. of inputs say $x(t) = 3x_1(t) + 4x_2(t) = 3t + 4\sin t$

Now. Output: $T\{x(t)\} = 2(3t + 4\sin t) = 6t + 8\sin t$ — ①

Now, Applying Superposition,

$$3C_1(t) + 4C_2(t) = 3(2t) + 4(2\sin t) = 6t + 8\sin t \text{ — ②}$$

from ① & ②, System is Linear.

eg. $C(t) = x(t) + \alpha$

Let x_1 & x_2 be the 2 input signals

Hence, $C_1(t) = x_1(t) + \alpha$

$$C_2(t) = x_2(t) + \alpha$$

Now let input $x = a_1 x_1 + a_2 x_2$

hence $C(t) = a_1 x_1 + a_2 x_2 + \alpha$ — ①

But for Superposition

$$C = a_1(x_1 + \alpha) + a_2(x_2 + \alpha) = a_1 x_1 + a_2 x_2 + \alpha(a_1 + a_2) \text{ — ②}$$

$\therefore ① \neq ② \Rightarrow$ Superposition fails $\Rightarrow$ Non-Invariant.

Note:

$$T\{x(t)\} = 2x(t)$$

Note: If system just scales, adds, delays, then it is often linear.

If system involves shifts by constants (like $+ \alpha$) powers (like sq, cube) exponentials, log, then it is often non-linear.

→ Significance of Linear Time Invariant System

- ① Using superposition any complicated signal can be broken down into simpler basic signals. Also, LTI systems follow linearity.
- ② General signals can be represented as L.C of delayed impulses which enables usage of time invariance property.